

Field Demonstration of QKD–Classical Coexistence with Real-Time Encryption over Multi-Core Fiber

Ibrahim M. Ramadan⁽¹⁾, Giammarco Di Sciullo⁽¹⁾, Qi Wu^(1,2), Domenico Ribezzo⁽¹⁾, Divya A. Shaji⁽¹⁾, Lucas A. Zischler⁽¹⁾, Ruben S. Luís⁽³⁾, Paolo Serena⁽⁴⁾, Chiara Lasagni⁽⁴⁾, Alberto Bononi⁽⁴⁾, Tetsuya Hayashi⁽⁵⁾, Alessandro Gagliano⁽⁶⁾, Paolo Martelli⁽⁶⁾, Alberto Gatto⁽⁶⁾, Paola Parolari⁽⁶⁾, Pierpaolo Boffi⁽⁶⁾, Andrea Marotta⁽¹⁾, Carlo Centofanti⁽¹⁾, Fabio Graziosi⁽¹⁾, Antonio Mecozzi⁽¹⁾, Cristian Antonelli⁽¹⁾

⁽¹⁾ University of L'Aquila, L'Aquila 67100, Italy, ibrahimmohamed.ramadan@univaq.it.

⁽²⁾ Photonics Research Institute, The Hong Kong Polytechnic University, Hong Kong, China.

⁽³⁾ National Institute of Information and Communications Technology, Tokyo 184-8795, Japan.

⁽⁴⁾ Department of Engineering and Architecture, Università degli Studi di Parma, 43124 Parma, Italy.

⁽⁵⁾ Sumitomo Electric Industries, Ltd., Sakae-ku, Yokohama 244-8588, Japan.

⁽⁶⁾ Department of Electronics, Information and Bioengineering, Politecnico di Milano, 20133 Milano, Italy.

Abstract

We report the first field trial of quantum key distribution coexisting with classical signals over a field-deployed multi-core fiber link, achieving 106.7 Tb/s classical transmission alongside real-time network-layer encryption at 132 Gb/s, with 63 μ s latency overhead and key-refresh time interval of 15 seconds.

Introduction

Unlike classical public key cryptography, quantum key distribution (QKD) enables distant parties to share encryption keys whose security is derived from the laws of quantum physics, making it resilient against any level of computing power, including future quantum computers [1, 2].

Progress has been made in point to point QKD demonstrations, and commercial systems are now available. Coexistence of QKD with high-capacity classical traffic over the same installed fiber has also been explored [3–5]. A promising approach involves multi-core fibers (MCFs), originally developed for space-division multiplexed transmission [6–8]. By dedicating one core to QKD and the remaining cores to classical data, it is possible to keep both signals within the C-band while limiting quantum noise from Raman scattering and four-wave mixing to acceptable levels [9, 10]. However, with the exception of [9, 10], previous experiments involving MCF were conducted in controlled laboratory environments [11–14], and none demonstrated real-time encryption of live user traffic.

The integration of QKD with real-time, high-speed data encryption over deployed infrastructures at rates and latencies compatible with commercial applications remains unproven, and the gap between lab-based validation and operational network deployment has yet to be explored.

In this work, we address these limitations through two key contributions. First, we demonstrate bidirectional transmission of 106.7 Tb/s full C-band classical traffic over 25.2 km of field-deployed uncoupled-core four-core fiber (UC-4CF) in L'Aquila, Italy, with one core reserved for QKD.

Unlike our previous work [9], which employed a prototypal QKD system, here we integrate a commercial QKD system, achieving a secret key rate of 1.7 kb/s under full C-band loading at 20 dBm launch power per core. Second, and most significantly, we demonstrate real-time encryption of live user traffic, securing up to 132 Gb/s within a 400 Gb/s WDM channel. The encryption introduces only 63 μ s additional latency and 2 μ s jitter compared to unencrypted transmission, remaining fully compatible with low-latency applications such as financial trading and cloud computing. This work represents the first field trial of practical QKD deployment in MFCs, proving that quantum-secured encryption can operate seamlessly within real high-speed networks without service degradation.

Experimental Setup & Encryption Architecture

The experiment was carried out over an UC-4CF deployed in the city of L'Aquila, Italy [15]. Despite the low inter-core crosstalk of -60 dB/km, quantum channel coexistence imposes constraints on system design [9, 16]. Specifically, classical signals in cores adjacent to the quantum core must counter-propagate relative to the quantum signal, whereas those in the diagonally opposite core may propagate in either direction; however, co-propagation is necessary to support bidirectional transmission. As shown in Fig. 1, core 4 (C4) is used for the quantum channel. The classical signal in the diagonally opposite core 1 (C1) co-propagates with the QKD signal from Alice to Bob, while cores 2 and 3 (C2 and C3) carry counter-propagating classical signals. The relevant fiber parameters are reported in [15, 16].

The QKD system, commercialized by an Italian

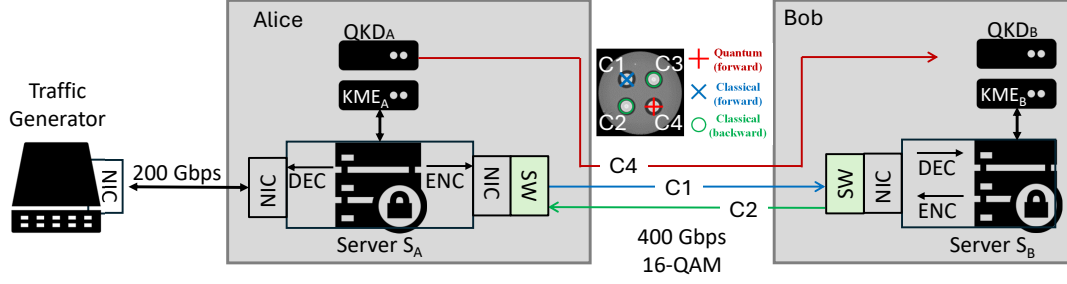


Fig. 1: Architecture for real-time encryption and decryption of classical data streams using keys from the QKD system. NIC: network interface card; ENC: encryption; DEC: decryption; KME: key management entity.

company [17], comprises a quantum-state transmitter (Alice), receiver (Bob), and a Quantum Key Management Entity (QKME). It implements the 3-state efficient BB84 protocol with one decoy state and time-bin encoding [18]. The Z basis is used for key generation and transmitted more frequently, while the X basis serves solely for security verification. The QKME is fully compliant with ETSI GS QKD 004, 014, 015, and 018 standards [19–22], and provides keys via REST APIs.

Full C-band bidirectional classical transmission over the 25.2 km UC-4CF is achieved using an experimental setup consistent with that detailed in [10]. Specifically, a test-band is generated using four 100-GSa/s digital-to-analog converters, a dual-polarization IQ modulator, and an external cavity laser, tunable from 1529.07 nm to 1566.21 nm. Spectrally-flattened amplified spontaneous emission (ASE) noise is used to emulate a fully populated C-band at launch power levels ranging between 10 dBm and 20 dBm per core. Figure 2(a) shows the transmitted spectrum, with a 100-GHz notch at the quantum channel centered at 1545.32 nm. A second 100-GHz notch centered at 1552.66 nm allows accommodating real-time transmission with small form-factor pluggable (SFP) transceivers, as discussed later in this section. Figure 2(b) shows the per-core data rate for all 183 WDM channels, as well as the aggregate data rate estimated using generalized mutual information (GMI) and after forward-error correction (FEC) decoding, at a launch power of 20 dBm per core. The FEC scheme uses codes from the digital video broadcasting (DVB-S2) standard in conjunction with code-rate puncturing to achieve error-free transmission [23]. The secure key rate (SKR) is measured at classical launch power levels of 10 and 20 dBm per core, as shown in Fig. 2(c). The time-averaged SKR decreases from approximately 2.15 kb/s to 1.75 kb/s with increasing launch power. The measured mean SKR values show good agreement with the predictions of the model presented in [9].

The architecture for real-time encryption is shown in Fig. 1. Here, the transmitter and receiver are implemented using commercial OpenZR+ capable Quad Small Form-factor Pluggables [24],

and the generated signal is combined with flattened ASE noise to obtain the spectrum shown in Fig. 2(a). The system handles 400 Gb/s transmission and is built around two main components: the QKD system, which generates the cryptographic keys, and two high-speed servers, S_A (Alice) and S_B (Bob), each equipped with an NVIDIA BlueField-3 SmartNIC and interconnected through a network switch. Applications on each server control 200 Gb/s of traffic generated by a traffic generator, while the remaining 200 Gb/s are generated by the switches. Access to the QKME is authenticated via certificates.

Encryption and decryption are implemented in C using the Data Plane Development Kit (DPDK) [25], which processes packets entirely in user space, bypassing the operating system kernel stack and eliminating the overhead of context switching and system calls. Combined with huge-page memory allocation and CPU core isolation, this sustains throughput exceeding 200 Gb/s while performing real-time cryptographic operations.

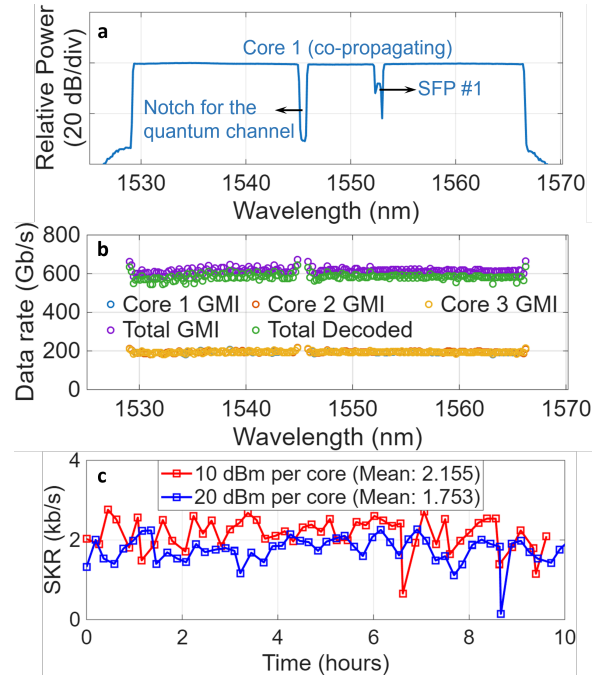


Fig. 2: (a) Spectrum of the C-band classical signal transmitted in core 1. (b) Per-core and aggregate data rates estimated using GMI and after FEC-decoding. (c) SKR versus time for classical launch power levels of 10 dBm and 20 dBm per core.

The system uses the Lightweight Encryption Algorithm (LEA) [26], a block cipher introduced by the Korea Internet and Security Agency in 2013. LEA operates exclusively on ARX (Addition, Rotation, XOR) operations on 32-bit words, making it significantly more efficient than AES on general-purpose processors. For LEA-128, key scheduling expands a 128-bit master key into 24 rounds of sub-keys, each comprising six 32-bit words. Encryption operates on 128-bit plaintext blocks divided into four 32-bit words (X_0, X_1, X_2, X_3), applying modular addition, XOR, and left bit rotations per round. Decryption reverses this process using the sub-keys in reverse order, applying right rotations, XOR, and modular subtraction.

Performance is further enhanced through Intel AVX-512 vector extensions, which enable simultaneous operations across 512-bit registers, reducing per-packet processing time under heavy traffic. NVIDIA flow steering technology is used at the network level to distribute traffic across CPU cores and isolate encrypted flows from unencrypted ones.

Results

The system performance was evaluated in terms of delay and jitter penalties introduced by real-time encryption across four scenarios: the first three consisted of 24-hour stress tests with a key refresh time interval of 30 seconds, while the fourth used a 15-second refresh time interval, each with varying splits between encrypted and unencrypted traffic. The results highlight the overhead introduced by the real-time encryption and confirm stable operation under field-deployed conditions.

In Scenario 1, traffic was evenly split between 100 Gb/s of encrypted and 100 Gb/s of forwarded flows. Encrypted traffic exhibited a mean end-to-end delay of 380.821 μ s, compared to 318.247 μ s for non-encrypted traffic, resulting in an encryption overhead of approximately 63 μ s. The mean jitter was 2.458 μ s for encrypted traffic and 0.304 μ s for non-encrypted traffic. In Scenario 2, the encrypted traffic was reduced to 50 Gb/s (with 150 Gb/s of forwarded traffic), with no appreciable change in the overhead introduced by the encryption process.

In Scenario 3, the system was pushed to its maximum encryption capacity, with 132 Gb/s of encrypted and 68 Gb/s of forwarded traffic. Figure 3(a) and (b) show a plot of the delay and jitter versus time, respectively. Encrypted traffic had a mean delay of 386.351 μ s against 324.036 μ s for non-encrypted traffic, around 63 μ s of overhead. A mean jitter of 2.307 μ s and 0.466 μ s is observed for encrypted and non-encrypted traffic, respectively. In Scenario 4, the system was pushed to its maximum key refresh rate capacity, with a refresh time interval of 15 s, and an encryption rate of 102 Gb/s. The results in terms of delay and

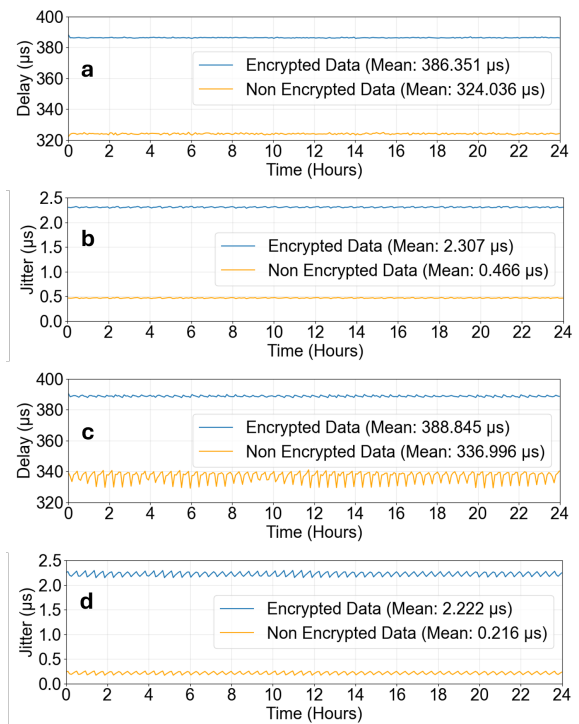


Fig. 3: Round-trip delay (a,b) and jitter (c,d) for encrypted and non-encrypted traffic under different scenarios. (a,b) Scenario 3: 132 Gb/s encrypted, 68 Gb/s forwarded, and 30 seconds key refresh time interval; (c,d) Scenario 4: 102 Gb/s encrypted and 97 Gb/s forwarded.

jitter versus time are shown in Fig. 3(c) and (d), respectively. In this case, the encryption overhead is approximately 52 μ s, with a jitter increase of about 2 μ s. Note that, given the achieved key refresh rate, the system encrypts 191.25 GB of data per key, which is substantially below the amount considered secure for encryption with a single 128-bit key under the LEA-128 protocol [27].

Across all scenarios, the encryption overhead varies by only approximately 10 μ s, and the jitter by only 0.7 μ s. This confirms that LEA-128 with AVX-512 acceleration introduces a practically fixed latency, regardless the encryption rate.

Conclusion

We have experimentally demonstrated the coexistence QKD, based on a commercial system, with 106.7 Tb/s of classical traffic over a field-deployed 25.2 km multi-core fiber link. We showcased record real-time encryption at the network layer of a 132-Gb/s live data stream with a key refresh time interval of 30 s, and a 102-Gb/s live data stream with a key refresh time interval of 15 s, within a 400-Gb/s channel in a full C-band configuration, using keys extracted from the QKD system. The hardware-based encryption introduced less than 63 μ s of end-to-end latency and below 3 μ s jitter, compared to unencrypted transmission. These results pave the way for scalable and quantum-safe optical networks in future high-capacity multi-core fiber infrastructures.

Acknowledgements

This work was funded by the European Union - Next Generation EU under the Italian National Recovery and Resilience Plan (NRRP), Mission 4, Component 2, Investment 1.3, CUP B53C22003970001, partnership on “Telecommunications of the Future” (PE000000001 - program “RESTART”), and funded by the European Union (Grant Agreement 101120422 and 101072409). Views and opinions expressed are, however, those of the author(s) only and do not necessarily reflect those of the European Union or REA. Neither the European Union nor the granting authority can be held responsible for them.

References

- [1] A. A. Hajomer, I. Derkach, R. Filip, U. L. Andersen, V. C. Usenko, and T. Gehring, “Continuous-variable quantum passive optical network”, *Light: Science & Applications*, vol. 13, no. 1, p. 291, 2024.
- [2] A. A. Hajomer, I. Derkach, N. Jain, H.-M. Chin, U. L. Andersen, and T. Gehring, “Long-distance continuous-variable quantum key distribution over 100-km fiber with local local oscillator”, *Science Advances*, vol. 10, no. 1, eadi9474, 2024.
- [3] J.-N. Niu, Y.-M. Sun, C. Cai, and Y.-F. Ji, “Optimized channel allocation scheme for jointly reducing four-wave mixing and Raman scattering in the DWDM-QKD system”, *Applied optics*, vol. 57, no. 27, pp. 7987–7996, 2018.
- [4] R. Wang et al., “End-to-end quantum secured inter-domain 5G service orchestration over dynamically switched flex-grid optical networks enabled by a q-ROADM”, *Journal of Lightwave Technology*, vol. 38, no. 1, pp. 139–149, 2019.
- [5] A. A. Hajomer, I. Derkach, V. C. Usenko, U. L. Andersen, and T. Gehring, “Coexistence of continuous-variable quantum key distribution and classical data over 120-km fiber”, *arXiv preprint arXiv:2502.17388*, 2025.
- [6] T. Matsui, P. L. Pondillo, and K. Nakajima, “Weakly coupled multicore fiber technology, deployment, and systems”, *Proceedings of the IEEE*, vol. 110, no. 11, pp. 1772–1785, 2022.
- [7] X. Wang et al., “Field trial of $7 \times 89 \lambda \times 256$ Gb/s C-band classical/CVQKD co-existence transmission over 7-core fiber”, in *2023 Asia Communications and Photonics Conference/2023 International Photonics and Optoelectronics Meetings (ACP/POEM)*, IEEE, 2023, pp. 1–4.
- [8] M. Zahidy et al., “Practical high-dimensional quantum key distribution protocol over deployed multicore fiber”, *Nature Communications*, vol. 15, no. 1, p. 1651, 2024.
- [9] Q. Wu et al., “Integration of quantum key distribution and high-throughput classical communications in field-deployed multi-core fibers”, *Light: Science & Applications*, vol. 14, no. 1, p. 274, Aug. 2025. [Online]. Available: <https://www.nature.com/articles/s41377-025-01982-z>.
- [10] Q. Wu et al., “Demonstration of quantum key distribution coexisting with 110-tb/s classical transmission over multicore fibers”, in *European Conference on Optical Communication (ECOC)*, *Corresponding author, †These authors contributed equally, IEEE, Sep. 2025.
- [11] J. Dynes et al., “Quantum key distribution over multicore fiber”, *Optics express*, vol. 24, no. 8, pp. 8081–8087, 2016.
- [12] D. Bacco et al., “Boosting the secret key rate in a shared quantum and classical fibre communication system”, *Communications Physics*, vol. 2, no. 1, p. 140, 2019.
- [13] E. Hugues-Salas et al., “11.2 Tb/s classical channel coexistence with DV-QKD over a 7-core multicore fiber”, *Journal of Lightwave Technology*, vol. 38, no. 18, pp. 5064–5070, 2020.
- [14] J.-Q. Geng et al., “Integration in the C-band between quantum key distribution and the classical channel of 25 dBm launch power over multicore fiber media”, *Optics Letters*, vol. 47, no. 12, pp. 3111–3114, 2022.
- [15] T. Hayashi et al., “Field-deployed multi-core fiber testbed”, in *2019 24th OptoElectronics and Communications Conference (OECC) and 2019 International Conference on Photonics in Switching and Computing (PSC)*, IEEE, 2019, pp. 1–3.
- [16] Q. Wu et al., “Co-existence of quantum key distribution and classical transmission in a field-deployed uncoupled-core four-core fiber”, in *ECOC 2024; 50th European Conference on Optical Communication*, VDE, 2024, pp. 1355–1358.
- [17] [Online]. Available: <https://www.qticompany.com/products/#quell-x>.
- [18] D. Ribezzo et al., “Deploying an inter-European quantum network”, *Advanced Quantum Technologies*, vol. 6, no. 2, p. 2200061, 2023.
- [19] European Telecommunications Standards Institute (ETSI), “Quantum key distribution (qkd); application interface”, ETSI, Tech. Rep. GS QKD 004 V2.1.1, Aug. 2020, Group Specification (GS).
- [20] ETSI GS QKD 014 V1.1.1, “Quantum Key Distribution (QKD); Protocol and data format of REST-based key delivery API”, European Telecommunications Standards Institute, Standard, Dec. 2019.
- [21] ETSI GS QKD 015 V2.1.1, “Quantum Key Distribution (QKD); Control Interface for Software Defined Networks”, European Telecommunications Standards Institute, Standard, Apr. 2022.
- [22] European Telecommunications Standards Institute (ETSI), “Quantum key distribution (qkd); orchestration interface for software defined networks”, ETSI, Tech. Rep. GS QKD 018 V1.1.1, Apr. 2022, Group Specification (GS).
- [23] D. S. Millar et al., “Design of a 1 tb/s superchannel coherent receiver”, *Journal of Lightwave Technology*, vol. 34, no. 6, pp. 1453–1463, 2016.
- [24] [Online]. Available: <https://smartoptics.com/product/so-tqsfdd4cczrp/>.
- [25] Linux Foundation, *Data Plane Development Kit (DPDK)*, 2015. [Online]. Available: <http://www.dpdk.org>.
- [26] S. Singh, P. K. Sharma, S. Y. Moon, and J. H. Park, “Advanced lightweight encryption algorithms for iot devices: Survey, challenges and solutions”, *Journal of Ambient Intelligence and Humanized Computing*, vol. 15, no. 2, pp. 1625–1642, 2024.
- [27] K. Bhargavan and G. Leurent, *On the practical (in-)security of 64-bit block ciphers: Collision attacks on HTTP over TLS and OpenVPN*, Cryptology ePrint Archive, Paper 2016/798, 2016. DOI: 10.1145/2976749.2978423. [Online]. Available: <https://eprint.iacr.org/2016/798>.